

A wave generation toolbox for the open-source CFD library: OpenFoam®

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SUMMARY

The open-source CFD library OpenFoam® contains a method for solving free surface Newtonian flows using the Reynolds averaged Navier–Stokes equations coupled with a volume of fluid method. In this paper, it is demonstrated how this has been extended with a generic wave generation and absorption method termed ‘wave relaxation zones’, on which a detailed account is given. The ability to use OpenFoam for the modelling of waves is demonstrated using two benchmark test cases, which show the ability to model wave propagation and wave breaking. Furthermore, the reflection coefficient from outlet relaxation zones is considered for a range of parameters. The toolbox is implemented in C++, and the flexibility in deriving new relaxation methods and implementing new wave theories along with other shapes of the relaxation zone is outlined. Subsequent to the publication of this paper, the toolbox has been made freely available through the OpenFoam-Extend Community. Copyright © 2011 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Tools for numerical modelling of surface water waves are in practical applications generally limited to Boussinesq-type (e.g. [1]) or statistical modelling of the wave field (e.g. [2]). Both of these methods are limited in their applications: as examples, they are unable to yield information on the distribution of turbulence in the surf zone or the magnitude of forces due to overturning waves on structures as a consequence of the depth-integrated nature of the methods and the fact that they rely on potential flow theory.

A considerable amount of freedom is gained by approaching the problem by using free surface modelling in the context of the Reynolds averaged Navier–Stokes equations. The extended freedom comes at the cost of an additional (vertical) dimension in the computational domain and large computational demands, typically requiring software, which can run in parallel. Today’s commercial CFD packages increase in cost with the number of applied processors, whereas the expenses for hardware are reasonable.

In recent years, the freely available CFD library OpenFoam® has gained popularity, and active communities are appearing within a wide range of fields, in which CFD is frequently used by practising engineers as well as researchers. Such coherent and focused community work is still in its infancy with respect to free surface flows for coastal, marine and maritime topics. A key element for coastal engineering studies currently lacking within OpenFoam is the ability to generate and

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absorb surface water waves in a flexible manner. The modelling of propagating water waves in OpenFoam has previously been studied by Morgan *et al.* [3]; however, they suffer from the lack of outlet relaxation zones, and in the work by Afshar [4], the relaxation technique suffers from a requirement of highly refined computational meshes around the water surface in order to achieve an accurate description of the wave kinematics. Both of these limitations are addressed and solved in the present work.

Irrespective of these former efforts, the official versions are released without the possibility of generating waves or absorbing internally generated waves (the latter is a considerable deficiency for offshore and maritime engineers). The present paper will describe such a toolbox, which is made available to the coastal and maritime engineering communities. Additionally, the ability of the model, coupled with the standard volume of fluid (VOF) scheme in OpenFoam, is demonstrated to accurately model propagating and breaking waves. The reflection coefficient from an outlet is described.

The structure of this paper is as follows. The adopted numerical approach for solving the free surface flows are described in Section 2, in which a detailed account on the developed boundary conditions and relaxation zone technique for generation/absorption of waves will be given. In Section 3, the ability to model and absorb wave phenomena is demonstrated. The article is concluded with a discussion of the results and a discussion for possible improvements. Using this toolbox as a starting point, it can easily be applied by practising engineers, when detailed simulations of wave phenomena are required. It may also serve as a common reference frame for research purposes.

2. NUMERICAL METHODS

The following section will first briefly touch on the governing equations and afterwards consider the wave generation adjacent to inlet boundaries and the relaxation of waves at inlet and outlet boundaries.

2.1. Governing equations

The governing equations for the combined flow of air and water are given by the Reynolds averaged Navier–Stokes equations (e.g. [5, chap. 4])

$$\begin{aligned} \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot [\rho \mathbf{u} \mathbf{u}^T] = & -\nabla p^* - \mathbf{g} \cdot \mathbf{x} \nabla \rho \\ & + \nabla \cdot [\mu \nabla \mathbf{u} + \rho \boldsymbol{\tau}] + \sigma_T \kappa_\gamma \nabla \gamma \end{aligned} \quad (1)$$

coupled with the continuity equation for incompressible flows

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u} = (u, v, w)$ is the velocity field in Cartesian coordinates, p^* is the pressure in excess of the hydrostatic, ρ is the density, $\mathbf{g}$ is the acceleration due to gravity and μ is the dynamic molecular viscosity. $\rho = \rho(\mathbf{x})$ varies with the content of air/water in the computational cells. $\boldsymbol{\tau}$ is the specific Reynolds stress tensor

$$\boldsymbol{\tau} = \frac{2}{\rho} \mu_t \mathbf{S} - \frac{2}{3} k \mathbf{I}, \quad (3)$$

where μ_t is the dynamic eddy viscosity, $\mathbf{S}$ is the strain rate tensor ($1/2(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$) and k is the turbulent kinetic energy per unit mass. ∇ is $(\partial/\partial x, \partial/\partial y, \partial/\partial z)^T$, where $\mathbf{x} = (x, y, z)$ are the Cartesian coordinates. The last term in Equation (1) is the effect of surface tension, where σ_T is the surface tension coefficient and κ_γ is the surface curvature (see [6]). The surface tension coefficient between air and water at 20°C is 0.074 kg/s²; however, its presence will only have minor effects in civil engineering applications.

The equations are solved for the two immiscible fluids simultaneously, where the fluids are tracked using a scalar field γ . γ is 0 for air and 1 for water, and any intermediate value is a mixture of the two fluids. The distribution of γ is modelled by an advection equation

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot [\mathbf{u}\gamma] + \nabla \cdot [\mathbf{u}_r\gamma(1 - \gamma)] = 0 \quad (4)$$

similar to the method of Hirt and Nichols [7]. The last term on the left-hand side is a compression term, which limits the smearing of the interface, and $\mathbf{u}_r$ is a relative velocity. The method is developed by OpenCFD®, and it is documented in [8]. Briefly, the implementation is based on an explicit first-order time integration routine, and the stability of the advection equation (Equation (4)) is achieved by using a flux limiter on the divergence term.

Using γ , one can express the spatial variation in any fluid property, such as μ and ρ , through the weighting

$$\Phi = \gamma\Phi_{\text{water}} + (1 - \gamma)\Phi_{\text{air}}, \quad (5)$$

where Φ can be any such quantity.

Turbulence becomes important in the case of wave breaking. It is modelled using a $k - \omega$ closure model on the basis of the description by Wilcox [9,10], where k is the turbulent kinetic energy and ω is the characteristic frequency for the turbulence. The two additional advection–diffusion equations take the following form for ω

$$\begin{aligned} \frac{\partial \rho \omega}{\partial t} + \nabla \cdot [\rho \mathbf{u} \omega] &= \alpha \mathcal{P}_\omega - \beta \omega^2 \\ &+ \frac{\sigma_d}{\omega} \nabla k \cdot (\nabla \omega)^T + \nabla \cdot \left[\left(\mu + \sigma_\omega \frac{k}{\omega} \right) \nabla \omega \right] \end{aligned} \quad (6)$$

and for k

$$\begin{aligned} \frac{\partial \rho k}{\partial t} + \nabla \cdot [\rho \mathbf{u} k] &= \mathcal{P}_k - \beta^* k \omega \\ &+ \nabla \cdot [(\mu + \sigma^* \mu_T) \nabla k]. \end{aligned} \quad (7)$$

The dynamic eddy viscosity, μ_t , is defined as

$$\mu_t = \rho \frac{k}{\tilde{\omega}}, \quad \tilde{\omega} = \max \left[\omega, \frac{C_{\text{lim}}}{\sqrt{\beta^*}} \sqrt{2\mathbf{S} : \mathbf{S}} \right], \quad (8)$$

where $:$ is the double inner product. The closure coefficients are $\alpha = 0.4$, $\beta = 0.0708$, $\beta^* = 9/100$, $C_{\text{lim}} = 7/8$, $\sigma_\omega = 1/2$, $\sigma^* = 3/5$ and

$$\sigma_d = \begin{cases} 0 & , \quad \nabla k \cdot (\nabla \omega)^T \leq 0 \\ 1/8 & , \quad \nabla k \cdot (\nabla \omega)^T > 0 \end{cases}. \quad (9)$$

The standard value of α is $13/25$, but because of a modification to the production term (see succeeding discussions), it has been found by comparing undertow profiles (Section 3.3) that the above value yields better results.

The production terms, $\mathcal{P}_\omega$ and $\mathcal{P}_k$, in Equations (6) and (7), take a form different from those in [9]. Mayer and Madsen [11] showed through both linear stability analysis and from practical experience that turbulent kinetic energy is extracted from the potential flow (outside the surf zone), when using the standard formulation. Instead, they suggested the following formulation, which they showed did not suffer from these problems:

$$\mathcal{P}_k = \mu_T (\nabla \times \mathbf{u}) \cdot (\nabla \times \mathbf{u})^T, \quad \mathcal{P}_\omega = \frac{\omega}{k} \mathcal{P}_k. \quad (10)$$

Hence, instead of determining the production on the basis of the strain rate, the production is based on the rotation of the velocity field. Therefore, no production takes place in the potential part of the

flow. Without this modification, a build-up of turbulence is found similar to that reported by Mayer and Madsen [11].

OpenFoam® uses a finite volume discretisation on unstructured grids consisting of arbitrary convex polyhedrals. A thorough description of the discretisation schemes in both space and time can be found in [12]. The pressure–velocity coupling is solved using the PISO algorithm [13], where its practical implementation is discussed in [12]. In this work, an implicit Euler time stepping is used [14].

The results in the present work are all obtained by using OpenFoam v. 1.5-dev. The released toolbox, however, compiles on the more recent versions of OpenFoam (the versions 1.6, 1.6-ext and 1.7.1 have been tested).

2.2. Boundary conditions

Each computational cell adjacent to a boundary has a cell face in common with this boundary. In the following, this face will be referred to as the boundary face (see Figure 1). The boundary face can be in three states, namely, completely submerged in water (wet), completely above the water surface (dry), and being intersected by the water surface (interface).

For the dry faces, the boundary conditions are

$$\mathbf{n} \cdot \nabla p^* = 0, \quad \mathbf{u} = \mathbf{0}, \quad \gamma = 0, \quad (11)$$

where $\mathbf{n}$ is the boundary unit normal vector. This corresponds to having a rigid wall extending down to the water surface. The boundary conditions for the wet faces are given analytically according to the chosen potential wave theory, which is evaluated at the face centres.

With respect to the boundary faces, which are intersected by the free surface, η , the intersection is computed as sketched in Figure 2. The two intersection points, I and II, are computed, and a local linear approximation to the real surface is defined, η^* . η^* forms a closed and wet polygon together with the wet sides of the boundary face. The corresponding wet area, A_w , and wet centre, $\mathbf{c}_w$, can be computed using simple geometrical methods. On the basis of the sub-division of the boundary face, γ at the boundary is specified as A_w/A_f , where A_f is the area of the boundary face. $\mathbf{u}$ and $\mathbf{n} \cdot \nabla p^*$ are evaluated from potential theory at $\mathbf{c}_w$ and applied onto the boundary face.

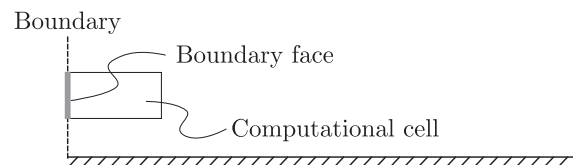


Figure 1. A definition sketch of a computational cell adjacent to the boundary and its boundary face.

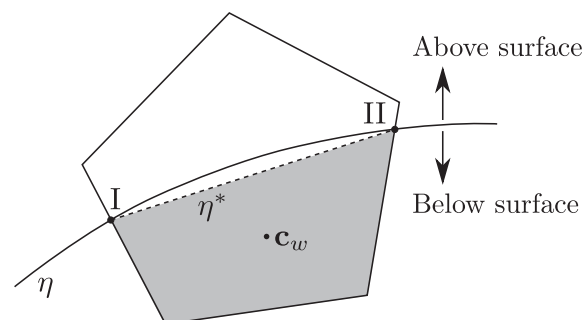


Figure 2. A sketch of the intersection between the surface elevation and a boundary face. The shaded part of the boundary face is the wet area, A_w . I and II are the two intersection points between the boundary face and the surface, η .

This approach is preferable to a typically used approach, where the face is either wet or dry based solely on the location of the face centre relative to η , as the latter approach resulted in spurious oscillations on the water surface. By applying the new approach, the spurious oscillations are avoided because of the smoothness, which is introduced at the interface.

At the upper boundary, an atmospheric boundary condition is used, which means that the air and the water are allowed to flow out, whereas only air is allowed to flow in. Therefore, the upper limit of the computational domain should be sufficiently far away to avoid a sink of water at the upper boundary.

In Sections 3.1 and 3.2, only a few wavelengths are considered; thus, the boundary layer dissipation is negligible (see [15, p. 50]). The use of a slip boundary condition is therefore justifiable at the bottom and is utilised. For the breaking wave case in Section 3.3, wall functions are used for the turbulent quantities in a fashion similar to that described by Nichols and Nelson [16]. The near wall values of ω and k are prescribed by (see e.g. [9, p. 160])

$$\omega_{nw} = \frac{u_f}{\sqrt{\beta^*} \kappa \Delta y}, \quad k_{nw} = \frac{u_f^2}{\sqrt{\beta^*}} \quad (12)$$

where $\kappa (= 0.40)$ is van Karman's constant, u_f is the magnitude of the friction velocity vector and Δy is the discretisation perpendicular to the wall. u_f is found by

$$\frac{\|\mathbf{u}_{nw,\tau}\|_2}{u_f} = \frac{1}{\kappa} \ln \frac{30\Delta y}{k_N} \quad (13)$$

where $\mathbf{u}_{nw,\tau}$ is the tangential part of the near wall velocity and k_N is Nikuradse's roughness height. Using a near wall resolution of $\Delta y = \mathcal{O}(k_N)$ has the drawback that the numerical evaluation of the velocity gradient, and hence the vertical momentum exchange, is underestimated. For example, in a channel flow with a constant energy gradient, the discharge will increase with increasing Δy , if no measures are taken. The method to achieve the correct momentum exchange is to evaluate

$$\rho u_f^2 = (\mu + \tilde{\mu}_t) \frac{\delta \|\mathbf{u}_{nw,\tau}\|_2}{\delta n} \quad (14)$$

where n is a coordinate perpendicular to the wall, δ is the numerical differentiation, and u_f is known from Equation (13). An artificial eddy viscosity, $\tilde{\mu}_t$, is used to fulfil Equation (14), and it is applied to the boundary. This yields a constant vertical momentum exchange at the bed as a function of Δy .

2.3. Relaxation zones

Relaxation zones are implemented to avoid reflection of waves from outlet boundaries and further to avoid waves reflected internally in the computational domain to interfere with the wave maker boundaries. The former obviously contaminates the results, and the latter is found to create discontinuities in the surface elevation at the wave making boundary, which leads to divergent solutions.

The present relaxation technique is an extension to that of Mayer *et al.* [17]. A relaxation function

$$\alpha_R(\chi_R) = 1 - \frac{\exp(\chi_R^{3.5}) - 1}{\exp(1) - 1} \quad \text{for } \chi_R \in [0; 1] \quad (15)$$

is applied inside the relaxation zone in the following way

$$\phi = \alpha_R \phi_{\text{computed}} + (1 - \alpha_R) \phi_{\text{target}} \quad (16)$$

where ϕ is either $\mathbf{u}$ or γ . The variation of α_R is the same as in [18]. The definition of χ_R is such that α_R is always 1 at the interface between the non-relaxed part of the computational domain and the relaxation zone, as illustrated in Figure 3. Engsig-Karup [19] discusses the functional form of the relaxation function, α_R , in the context of discontinuous Galerkin methods. α_R must fulfil the requirements $\alpha_R(0) = 1$ and $\alpha_R^{(n)}(0) = 0$. This requirement is in the present case fulfilled up to

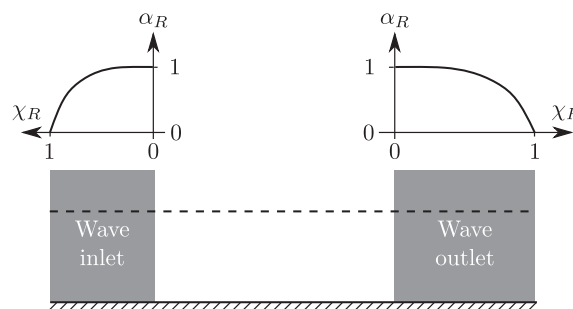


Figure 3. A sketch of the variation of $\alpha_R(\chi_R)$ for both inlet and outlet relaxation zones.

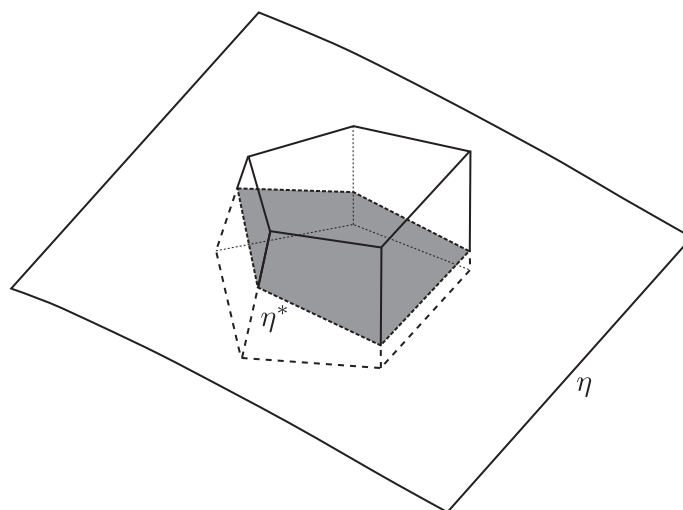


Figure 4. A sketch of the intersection between a computational cell and the water surface, η ; the intersection plane, η^* , is hatched. This creates an upper and lower volume, being dry and wet, respectively.

$\alpha_R^{(3)}(0)$ and is found to be sufficient. The degree to which the requirement is fulfilled can be changed by modifying the exponent of 3.5 in Equation (15).

In Equation (16), γ_{target} in a computational cell is either 1 or 0 if the cell is completely wet or completely dry. For computational cells intersected by η , the value of γ_{target} is between these limits, and its evaluation is achieved via an approach similar to the one at the boundaries (see Figure 4). The analytical surface elevation as a function of space and time is known in the relaxation zone; hence, the intersection between the surface elevation and the edges of the computational cell can be found. This defines two closed volumes, where the sum of the volumes equals the volume of the computational cell. The ratio between the wet and the full volumes yields γ_{target} . Additionally, the centre of the wet part of the volume can be evaluated on the basis of the intersections, and at this point, $\mathbf{u}_{\text{target}}$ is evaluated and assigned to the computational cell. In outlet relaxation zones, $\mathbf{u}_{\text{target}} = 0$ m/s, and γ_{target} is based on the position of the still water level.

The boundary conditions and the relaxation zones have been implemented using object-oriented programming (C++), and the geometrical intersection routine is decoupled from the actual wave theory under consideration. The adoption of object orientated programming implies that an extension of the wave theories merely requires knowledge of the variation of $\mathbf{u}$, η , and p^* as a function of space and time; hence, additional wave theories can easily be added. At present, first-order, second-order, and fifth-order stokes wave theories are available for regular waves together with cnoidal theory and stream function theory. In addition, a method for an arbitrary combination of the regular theories is available.

Two shapes of the relaxation zone is implemented: (i) a rectangular shape, which can be arbitrarily orientated in the horizontal plane (see Figure 3 for a two dimensional version), and (ii) a circular ring (see Figure 5). Other geometrical shapes of the relaxation zone are easily implemented owing to the object-orientated programming approach.

An arbitrary number of relaxation zones can be defined, and the source code is made ready, such that each relaxation zone can obey different relaxation approaches.

Both the wave boundary condition and the relaxation zone technique have been made available through the OpenFoam-Extend Community (www.extend-project.de; <http://sourceforge.net/projects/openfoam-extend>), together with test cases, pre-processing utilities, and solvers. See Appendix B for details on the download procedure.

3. RANGE OF APPLICABILITY

A thorough validation of OpenFoam as a tool for the modelling of surface water waves is not the scope here; thus, the description will be limited to (i) a validation with the experimental data presented by Chapalain *et al.* [20], (ii) an analysis of the ability to absorb regular waves in the shallow water limit and (iii) an example where the modelling of breaking waves is considered.

Parameters with respect to spatial discretisation, and the used Courant criterion on the time step, is given in Table I.

Other validation cases have been carried out, and they do also support that OpenFoam is applicable to perform wave modelling with this toolbox. These validation cases are (i) use of a relaxation zone to compute a linear standing wave, (ii) shoaling and release of higher harmonics on and over

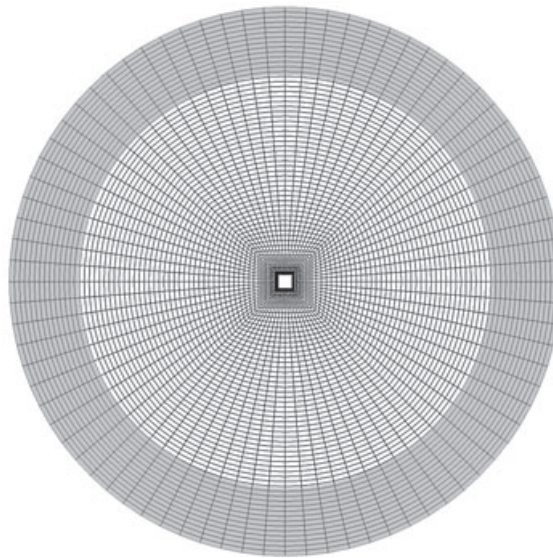


Figure 5. Top-down view on a computational domain with a square pile in the centre, where the circular relaxation zone is utilised. The grey area is the relaxation zone. The width of the relaxation zone is 10 m, and the diameter of the circle is 80 m.

Table I. Numerical discretisation parameters for all validation cases.

Case	Section	$\Delta x / \Delta z$ [m]	Δy at $y = 0$ [m]	Co. no. [–]	AR [–]	No. of cells $\cdot 10^3$
Triad interaction	3.1	0.030	0.0033	0.25	9.0	117
Reflection coefficient	3.2	0.030	0.0033	0.25	9.0	126–168
Wave breaking	3.3	Varying		0.25	1.0 / 2.0	107.2/53.6

AR = $\Delta x / \Delta y$ is the cell aspect ratio.

a submerged breakwater [21], and (iii) diffraction through a breakwater gap [22]. These validation cases are described and discussed in [23, Chap. 4].

3.1. Exchange of energy between the harmonics

The first example is the spatial change of a sinusoidal wave of finite wave height, which is generated at the inlet over a horizontal bed. The prescribed sinusoidal motion does not satisfy the complete non-linear wave problem; hence, bound higher-order harmonics are generated adjacent to the boundary. In order to fulfil the first-order theory at the boundary, however, spurious free higher harmonics are likewise created, which have amplitudes equal to their bound counterparts, but with opposite phase, such that they cancel one another at the inlet. Interaction between these harmonics leads to an energy transfer, which can be identified as beat lengths (see [24] for further details).

This phenomenon has been investigated experimentally by Chapalain *et al.* [20], and it is modelled in a wave flume of 39 m in length with a constant water depth, h , of 0.4 m. The sinusoidal wave is 0.084 m high with a period of 3.5 s, which corresponds to a wave length of 6.8 m ($kh = 0.37$). A relaxation zone of 13 m in length is 'installed' at the outlet. The horizontal discretisation is 0.03 m, and near the surface, the vertical discretisation is 0.0033 m. The computational mesh is coarsened towards the bed.

A comparison between the experimental results and the present numerical work is seen in Figure 6. The simulated results for the various harmonics are in good agreement with the experimental measurements. The harmonics are computed as outlined in Appendix A, Equation (18). The figure also depicts the amplitude on the different harmonics, if a stream function wave is generated at the inlet, that is, an exact propagating solution to the non-linear wave problem. This wave propagates along the flume with a constant shape, which serves to validate the method for generating stream function waves. Furthermore, this test qualitatively verifies the applicability of the adopted method for absorption of the waves at the outlet boundary, as no sign of reflections is apparent.

It appears that it is the first time that this benchmark case has been simulated with a VOF method.

3.2. Reflection from the outlet boundary

In the previous section, the outlet relaxation zone is qualitatively seen to be non-reflective. In the present section, the performance of the outlet relaxation zone will be quantified. The setup

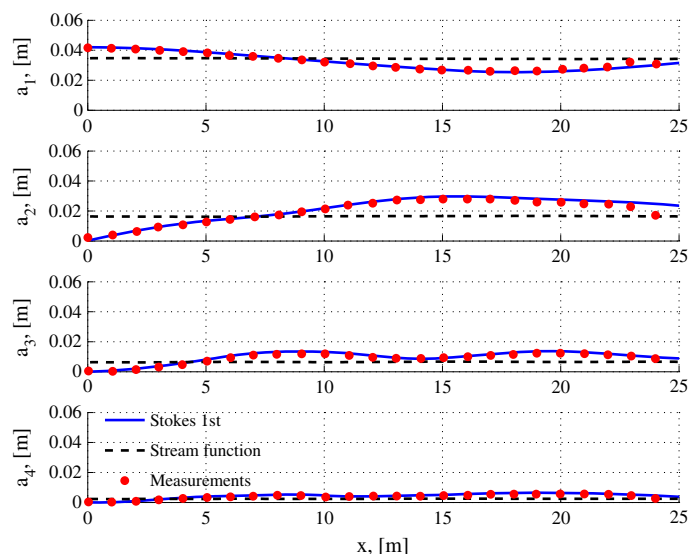


Figure 6. Comparison between the experimental data by Chapalain *et al.* [20] and the modelled results. Additionally, results applying stream function waves at the inlet are shown. a_1 , a_2 , a_3 and a_4 are the amplitude of the first four harmonic components including both free and bound harmonics.

from Section 3.1 is maintained, where the length of the outlet relaxation zone, λ , is varied relative to the wave length. A range of wave heights, H , is also considered. To ease the evaluation of the reflection coefficient, we based the generated waves on stream function theory, from which the reflection can be evaluated using Equation (20) in Appendix A.

The reflection coefficient in Figure 7 is defined as

$$R = \|(a_1^R; b_1^R)\|_2 / \|(a_1^I; b_1^I)\|_2 \quad (17)$$

with $\|\cdot\|_2$ being the 2-norm and only the first-order harmonics are used. The incident and reflected components are evaluated using Equation (20).

In Figure 7(A), the ratio between the propagating wave amplitude of the first harmonic and the amplitude of the reflected wave is plotted as a function of the ratio λ/L . It is seen that the performance is generally good, especially for relaxation zones having λ/L equal to or larger than 1. For increasing H/h , the reflection increases for a given λ . In Figure 7(B), R is plotted as a function of $\lambda h/(LH)$, that is, normalised by the wave height to depth ratio. This shows that the lines collapse except for $H/h = 0.4$, in which case wave breaking occurs before the waves reach the absorbing relaxation zone. The wave breaking is induced by the magnitude of the cell aspect ratio, $AR(=9.0)$ (see Sections 3.3 and 4 for a discussion). It should be stressed that in these tests, $kh = 0.37$; hence, these results can be expected to be valid for waves within or near the shallow water limit. The magnitude of R corresponds to similar methods for a hybrid numerical beach [25] and a complex self-adaptive outlet boundary condition [26].

Tests have also been made with no relaxation on γ . A single test is shown in Figure 7, and it is found to result in a reflection coefficient of $\mathcal{O}(0.3)$, whereas $R = 0.005$, when relaxation on γ is performed.

3.3. Validation case for breaking waves

As a further demonstration, the applicability of OpenFoam for simulating wave breaking has been tested against the laboratory experiment by Ting and Kirby [27] in the case of spilling breakers on a plane slope (1:35). Waves are generated on a horizontal bed, where the still water depth is 0.4 m (see Figure 8). The waves are 0.125 m high and have a period of 2 s; stream function waves have

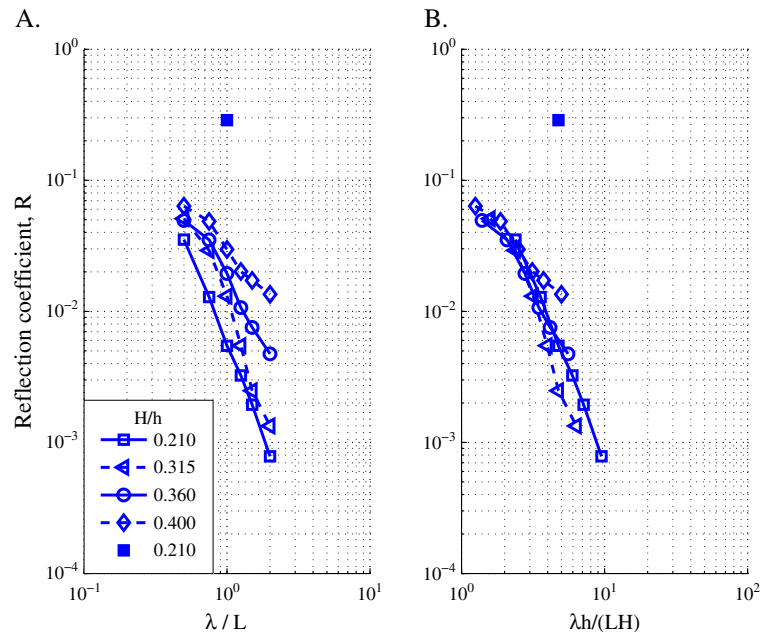


Figure 7. The reflection coefficient as a function of H/h . Filled square indicates simulation without relaxation on γ , whereas all other data include relaxation on γ . (A) Plotted as a function of λ/L and (B) $\lambda h/(LH)$.

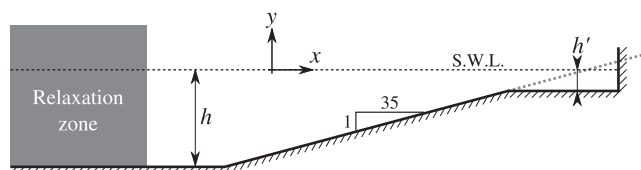


Figure 8. Layout of the computational domain for computing spilling breaking waves. The dashed grey line is the layout of the original physical experiment. The distance from origo to the toe is 0.7 m.

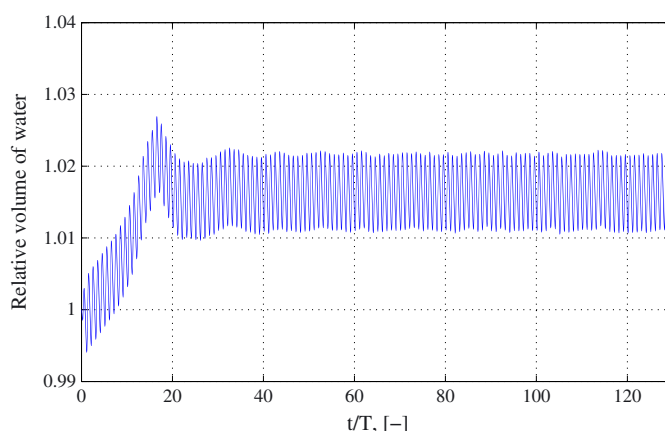


Figure 9. Relative variation in the total amount of water (V_γ) in the computational domain.

been used. A test has also been conducted with cnoidal waves, but no considerable difference is found. At the inlet, a relaxation zone 4 m in length is used to allow for a long simulation time, a requirement which is discussed by Bradford [28]. As shown in Figure 8, a cut-off of $h' = 0.01$ m is applied, where the swash zone was supposed to be. This is necessary, as the coupling between the VOF method and the momentum equation is not robust, when the water surface attaches on the bottom. The lack of robustness is caused by the high aspect ratio ($\mathcal{O}(100)$) along the bottom boundary, which is needed to have sufficiently fine near wall mesh resolution. The implications on the hydrodynamics is assumed to be limited, because (i) the cut-off is only 2.5% of the water depth at the inlet; hence, most of the wave energy is dissipated, and (ii) studies with a live bed [29] with and without a seawall at the still water level yielded minor differences in the evolving morphology; thus, the effect onto the hydrodynamics must have been equally small.

This test case has been considered with varying success by several other authors such as Lin and Liu [30], Bradford [28], Mayer and Madsen [11], Hieu *et al.* [31] and Christensen [32]. The results by Hieu *et al.* [31] are generally encouraging; however, their simulation time is limited to 25 s (i.e. $t/T = 12.5$) of warm-up, and the subsequent averaging was limited to five wave periods. This limited simulation time is common for the cited papers, and the reported results are at best quasi-steady. Contrary to the other cited papers, Hieu *et al.* [31] apply the defined donating region method (DDR by [33]) for the interface tracking, which is thought to have some interesting properties with respect to the breaking description (see Section 4 for a discussion). It must be noted, however, that they perform a mesh sensitivity analysis (see [31, figure 16]), and irrespectively of considerable changes with increasing resolution, no better resolved simulation is presented than that, which matches the experimental data nicely.

The bottom roughness is not explicitly stated in [27]; thus, it is assumed smooth, and a small roughness height, $k_N = 0.1$ mm, is used. The computational cells outside the boundary layer all have the same aspect ratio, AR , and an AR of 1 and 2 is considered. The averaging of the hydrodynamic properties is performed over 50 wave periods from $t/T = 80$ to $t/T = 130$. This choice is based on Figure 9, which depicts the volume of water in the computational domain ($V_\gamma = \sum_{\text{domain}} V_\gamma$) as

a function of time, where V is the volume of a computational cell. V_γ exhibits a transient behaviour, which describes the build-up of the wave setup. The results are at least affected by this transient behaviour up to $t/T = 40$. A problem with extracting data too early is the apparent lack of mass conservation, which is found in, for example, the simulation by Lin and Liu [30] (as noted by Christensen *et al.* [34]). It is worth noticing that no problems have been encountered even though the simulation time exceeds $100T$, which proves the method of inlet relaxation and absorption to be robust.

The surface elevation in the surf zone can be seen in Figure 10(A) ($AR = 1$) and (B) ($AR = 2$). There is a very noticeable difference between the two simulations, as the wave breaking occurs much earlier compared with the measurements for $AR = 2$, whereas the comparison is good for $AR = 1$. Several mesh resolutions have been tested, and the location of the break point is persistent for both $AR = 1$ and $AR = 2$ even though finer meshes are used.

The premature breaking for $AR = 2$ is thought to be explained by the nature of the VOF method. If the γ content is larger than 0, there will be a flux of γ over the cell faces irrespective of the actual location of the water surface. As soon as water enters a computational cell from the left and the cell has a large aspect ratio (> 1), there will be a flux of γ across the right face. This will lead to an excessive amount of water being advected horizontally and too little vertically. The combination of an excessive amount of horizontal mass transport and a decreasing horizontal velocity with the downward vertical distance from the wave crest leads to an artificial steepening of the wave front. This eventually causes the appearance of premature wave breaking.

The resulting undertow profile is depicted in Figure 11, and the comparison is generally favourable for $AR = 1$. The considerable discrepancy for the measuring positions F, G and H is explained by the too rapid surface elevation decay in the inner surf zone, which cause a steeper wave setup. In the case of $AR = 2$, it is seen that the undertow profiles differs from those with $AR = 1$, and

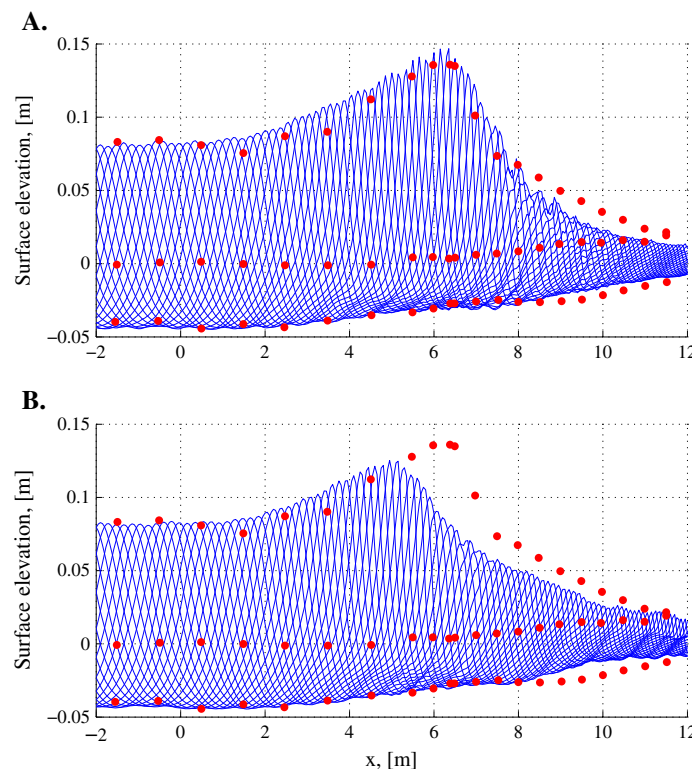


Figure 10. Measured [27] and simulated surface elevation in the surf zone in the case of spilling breakers. Lines are the ensemble average of the surface elevation for 20 phase over a wave period. (A) $AR = 1$. (B) $AR = 2$.

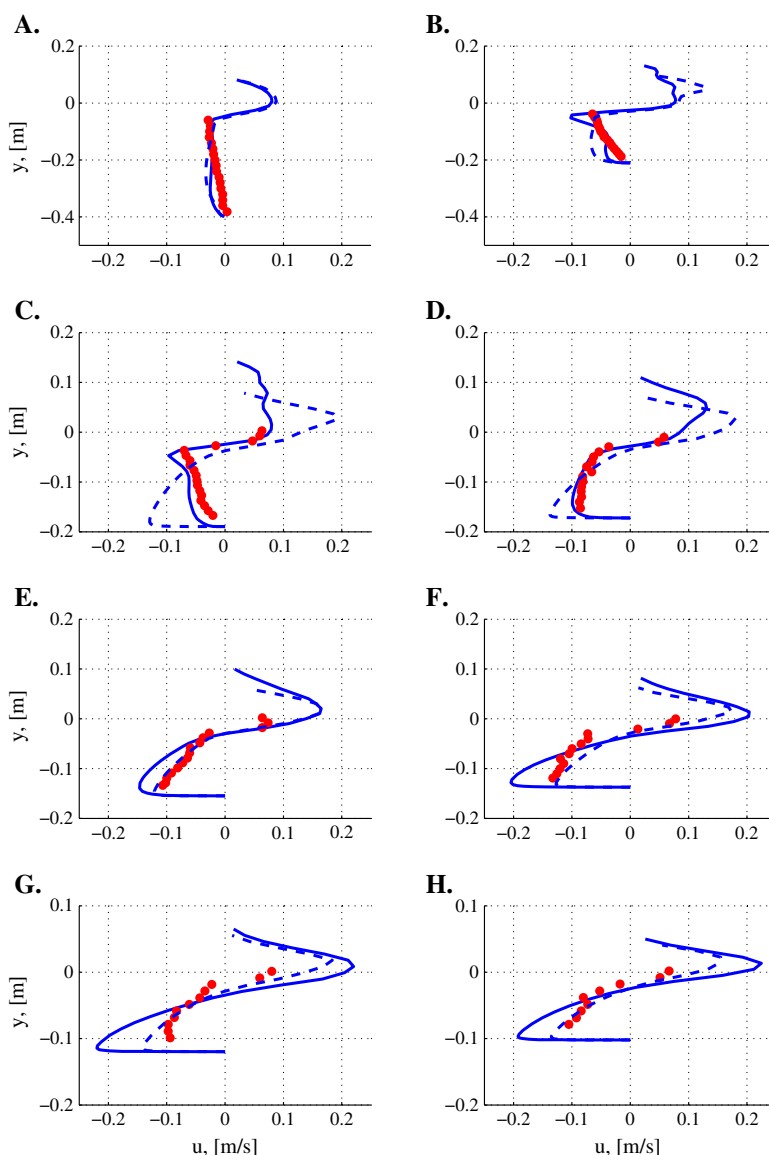


Figure 11. Measured [27] and simulated undertow profile at (A) $x = -1.265$ m, (B) $x = 5.945$ m, (C) $x = 6.665$ m, (D) $x = 7.275$ m, (E) $x = 7.885$ m, (F) $x = 8.495$ m, (G) $x = 9.110$ m, (H) $x = 9.725$ m. (Circles) Experiment. (Continuous lines) $AR = 1$. (Dashed lines) $AR = 2$.

the signature of the undertow with large flow velocities at the bed is found farther offshore. This is a direct consequence of the premature wave breaking. The two solutions are nevertheless comparable.

4. DISCUSSION

The present work has demonstrated that it is possible to use OpenFoam as a framework for the modelling of surface water waves a VOF method. The wave relaxation methodology is demonstrated to yield reasonable reflection coefficients in the limit of shallow water theory, which are similar to values reported elsewhere in the literature [25,26]. Future work should consider the ability to absorb waves in the intermediate and deep water regimes, that is, evaluating $R(kh)$.

Furthermore, a novel feature is presented, where it is possible for the user to specify custom geometrical shapes for the relaxation zones. This is exemplified with a circular ring type relaxation zone

(Figure 5), which would be straight-forward applicable when studying, for example, the dependency of the direction of hydrodynamic forcing relative to the orientation of the structure(s) under consideration. The method should, however, be used with caution combined with surface water waves, as their dissipation is sensitive to the radial mesh discretisation, whereas simulations with a uniform current are less sensitive to this aspect.

It is the authors' view that along with the public release of this wave generation/absorption toolbox, a common basis for the coastal, marine and maritime communities is formed. Any development, when published, can then be distributed throughout academia and among practising engineers to the mutual benefit for all. As with any other software, there is some obvious extensions, which would make the tool applicable wider.

An example is that most engineering applications are invariant to the air flow induced by the wave motion, an air motion, which is presently computed in OpenFoam. It would be of interest to develop a two-phase solver using any type of interface tracking and exclude the air flow from the governing equations (see e.g. [35]). The benefit of this would be twofold. First, the velocities induced in the air does sometime dominate the Courant criterion, which control the magnitude of the time step. A simple approach to avoid the large velocities and reduce the computational time is suggested by Liu and Garcia [36], where they set the air velocities to $\mathbf{0}$ each and every time step. This approach has been tested for a stream function wave using the setup from Section 3.1, and the spatial evolution of the harmonic amplitudes is presented in Figure 12. It is obvious that this approach leads to excessive amounts of dissipation of wave energy over several wave lengths. The method, however, is still reasonably applicable [36], because only one wave length is considered, and their primary interest were the bed shear stress for local scour. Second, the height of the computational domain is often set conservatively to avoid a loss of water through the atmospheric boundary; however, all the computational cells are included in the solution of the governing sets of equations. If the number of cells are limited to those filled with water, the computational time will be reduced.

Another obvious improvement would be to consider the effect of the aspect ratio (AR). As was mentioned in Sections 3.2 and 3.3, unphysical breaking occurs if AR is not 1. It is argued that this is caused by the cell averaged representation of the colour function, γ , which is advected incorrectly. The results by Hieu *et al.* [31] for wave breaking are encouraging, and they are using a different approach for the surface capturing, namely, a method, where the distribution of γ inside the computational cell is defined. This is the method derived by Harvie and Fletcher [33], and the internal

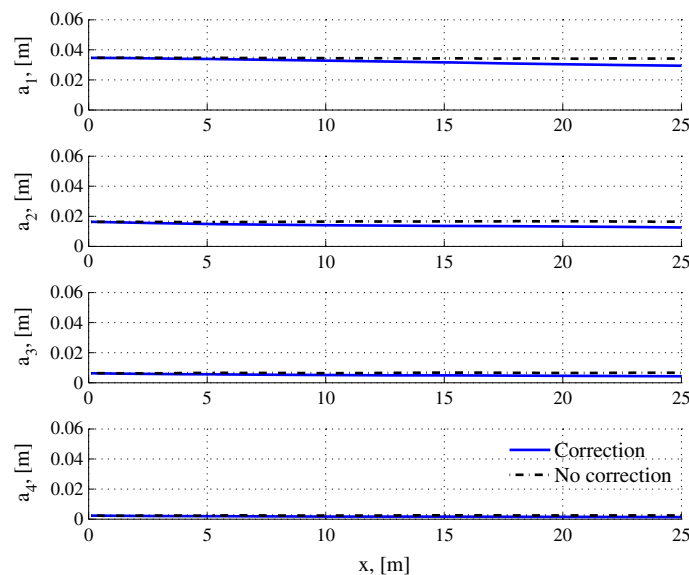


Figure 12. Dissipation of the wave amplitude when the velocities in the air is set to $\mathbf{0}$ at each time step. The wave length is 7 m.

distribution has a direct effect on the advection of γ . It is thus believed that an incorporation of the defined donation region method by Harvie and Fletcher [33] could improve the simulation of wave propagation/breaking even on meshes, where $AR \neq 1$.

The wave theories applied in the present work are not representative for natural occurring wave fields, which are irregular and have some degree of directional spreading. Focusing of waves by directional or phase focusing can cause the appearance of large waves and a corresponding large wave induced force [37], a force, which cannot be predicted using potential theory. In order to generate more naturally occurring incident waves, a coupling could be made to the spectral wave model SWAN as formulated by Booij *et al.* [2]. The spectral information could be translated to time series by, for example, the method of Sharma and Dean [38]. SWAN is similar to OpenFoam freely available under the GNU Public License.

5. CONCLUSION

A freely available wave generation toolbox is presented. The applicability of the toolbox to generate and absorb waves is demonstrated. It is also demonstrated that long-term simulations, which exceed 100 wave periods, can be achieved using this model in domains being only a few wave lengths long.

A couple of future improvements are suggested in the discussion, and the mutual benefit of a common framework across the disciplines of coastal, marine and maritime communities is underlined.

The toolbox is released as open-source under the GNU Public License through the OpenFoam-Extend Community.

APPENDIX A: EVALUATION OF HARMONIC AMPLITUDES

The surface elevation will be considered both in the time and the frequency domains. In the frequency domain, the method of extracting the energy distribution on frequencies is described here. As the generated waves are regular with period T , the frequency spectrum will have distinct peaks at $2\pi j/T$ for $j = 1, 2, \dots$ being 0 elsewhere. Hence, a Fourier transform will yield distorted results, as the energy will be smeared over frequency bins of finite width. Another method is adopted, where the optimal solution to the following over-determined set of equations

$$\eta_i = a_0 + \sum_{j=1}^N a_j \cos j\sigma t_i + b_j \sin j\sigma t_i \quad (18)$$

for

$$t_i = t_0, t_0 + \Delta t, \dots, t_0 + M\Delta t \quad (19)$$

at different positions in space is computed. Here, t_0 is chosen such that when multiplied with the group velocity, it will be larger than the distance from the inlet to the sampling position. $\sigma = 2\pi/T$ is the cyclic frequency, a_j and b_j are the harmonic amplitudes and a_0 is the mean water level.

The formulation in Equation (18) does not distinguish between incident and reflected waves of a given frequency. For cases involving reflected waves, the above method is applied on two points in space simultaneously. The over-determined set of equations then takes the form

$$\begin{aligned} \eta_{i,1} &= a_0 + \sum_{j=1}^N a_j^I \cos j(\sigma t_i - kx_1) + b_j^I \sin j(\sigma t_i - kx_1) \\ &\quad + a_j^R \cos j(\sigma t_i + kx_1) + b_j^R \sin j(\sigma t_i + kx_1) \\ \eta_{i,2} &= a_0 + \sum_{j=1}^N a_j^I \cos j(\sigma t_i - kx_2) + b_j^I \sin j(\sigma t_i - kx_2) \\ &\quad + a_j^R \cos j(\sigma t_i + kx_2) + b_j^R \sin j(\sigma t_i + kx_2) \end{aligned} \quad (20)$$

This set of equations is solved for values of t_i as given in Equation (19). The distance $x_2 - x_1$ is so small that it can be assumed that the wave amplitudes are altered by neither energy exchange nor numerical diffusion. Here, the superscripts I and R stand for incident and reflected wave components, respectively. k is the wave number and reflects the assumption that all harmonics are bound. The method in Equation (20) bears resemblance to that described by Liu and Yue [39], where the main difference is that they filter the surface elevation signal to leading order before computing the leading order harmonics coefficients.

APPENDIX B: DOWNLOAD INSTRUCTIONS

Instructions on how to download and install the toolbox can be found at <http://openfoamwiki.net/index.php/Contrib/waves2Foam>. This website will be maintained with the most recent developments and potential changes in download instructions.

The toolbox is made available within the OpenFoam-Extend Community (www.extend-project.de; <http://sourceforge.net/projects/openfoam-extend>), and it is released under a GNU Public License.

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